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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

# Mental Health Prediction using Human Emotions with Swin Transformer


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**PWCS22B01**

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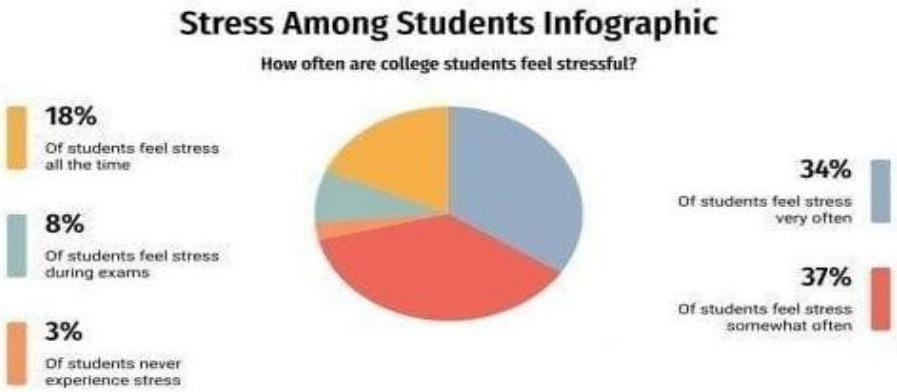
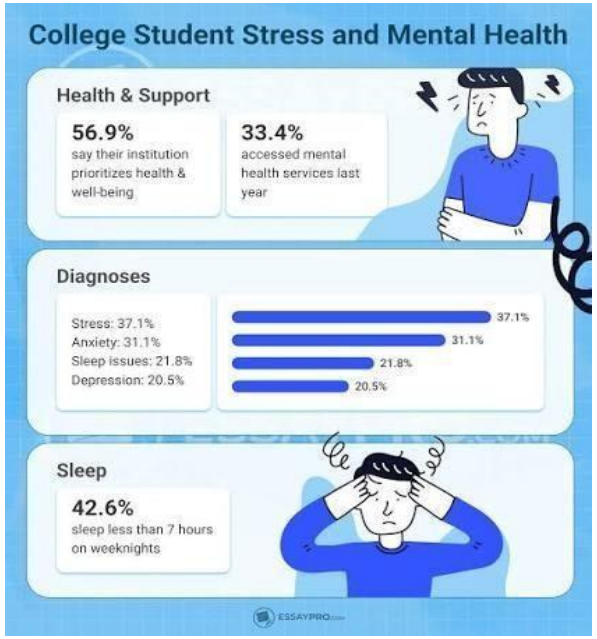
# INTRODUCTION

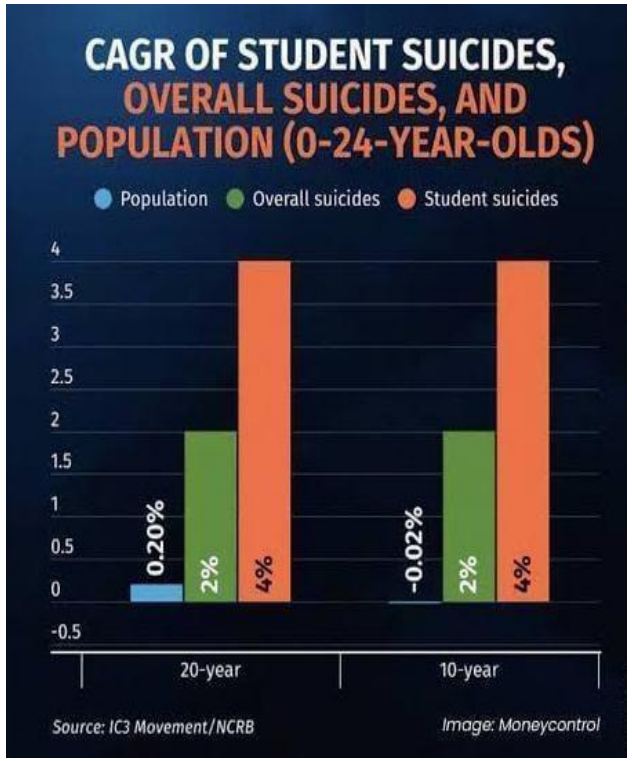
- Mental health refers to a person's emotional, psychological, and social well-being, influencing how they think, feel, and behave in daily life. It affects decision-making, stress handling, learning, and relationships.
- In India, mental health among students has become a growing concern. According to a 2025 India Today survey, nearly 3 in 5 students (60%) in major Indian cities report symptoms of anxiety, depression, or stress, and about 70% experience moderate to severe academic pressure.
- A UNICEF India report (2024) highlights that 1 in every 7 adolescents aged 15–24 suffers from mental health issues, often due to academic expectations, family stress, and competition.

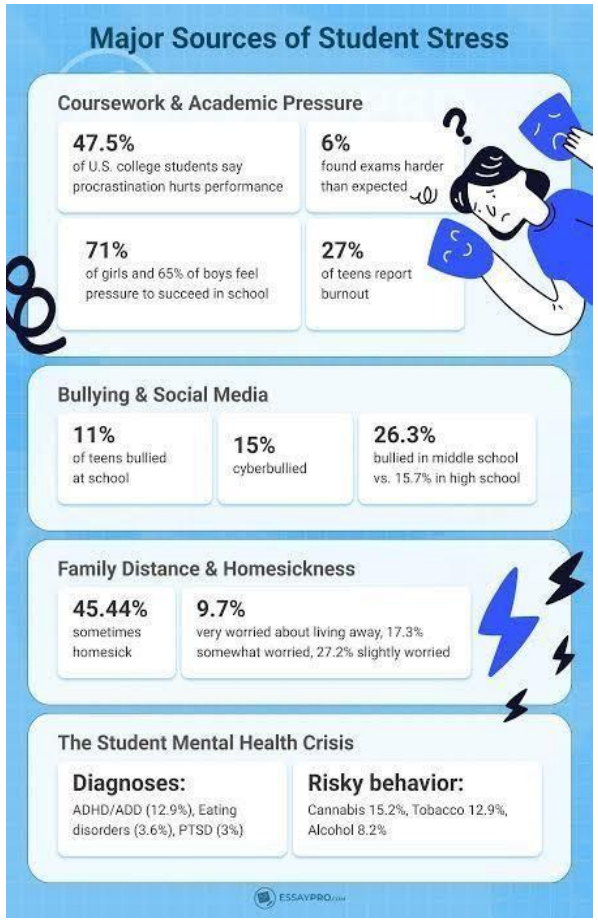


# INTRODUCTION

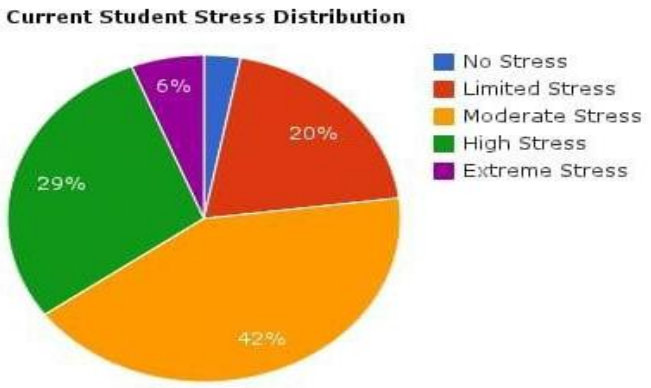
- Despite rising awareness, mental health problems often go undiagnosed because students hesitate to share their emotions, and institutions lack early detection systems.
- Poor mental health among students often arises from multiple factors such as academic overload, examination stress, comparison with peers, lack of proper rest, social media press, poor mental health among students often arises from factors like academic pressure, fear of failure, lack of rest, excessive social media use, and limited emotional support, which together lead to stress and emotional imbalance.
- If left unaddressed, these issues can cause a decline in focus, motivation, and academic performance, along with emotional instability, social withdrawal, and, in severe cases, self-harm tendencies — highlighting the urgent need for early detection and support.

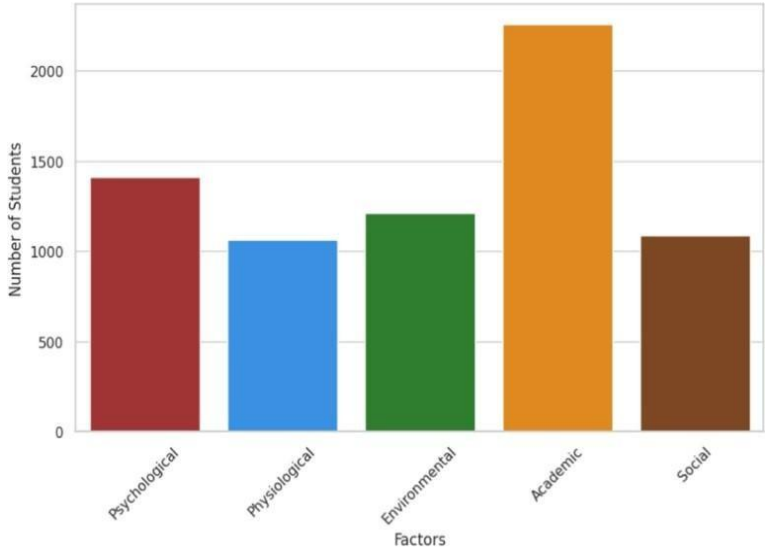


# INTRODUCTION

- Traditional mental health assessment methods like observation or surveys are often subjective, time-consuming, and fail to capture subtle emotional cues that indicate stress or declining well-being.
- With the rising stress levels among students, there is an urgent need for intelligent, non-invasive, and real-time systems that can support early detection of emotional distress.
- This project focuses on using facial emotion analysis to understand a person's mental state and generate a mental health score, promoting self-awareness and timely help-seeking behavior.



# LITERATURE SURVEY

| S.No | Year | Title | Method Used | Measures | Key Outcomes | Limitations |
|---|---|---|---|---|---|---|
| 1. | 2025 [1] | Real-Time Classroom Emotion Analysis Using Machine and Deep Learning for Enhanced Student Learning | Hybrid ML + Deep Learning (CNN, SVM, Ensemble models) | Accuracy, realtime performance | Real-time classroom emotion detection improves teaching insights and engagement | Requires highquality video input; model complexity |
| 2. | 2024 [2] | Mental Health and Stress Prediction Using NLP and Transformer-Based Techniques | Transformer based NLP (BERT, RoBERTa) | Accuracy, precision, recall | Transformer models effectively predict stress from text patterns | Limited to text input; contextual ambiguity |
| 3. | 2023 [3] | Vision Transformer With Attentive Pooling for Robust Facial Expression Recognition | Vision Transformer (ViT) + Attentive Pooling | Accuracy, robustness | Attentive pooling significantly enhances ViT performance for FER | Requires large datasets; computationally heavy |

# LITERATURE SURVEY

| S.No | Year | Title | Method Used | Measures | Key Outcomes | Limitations |
|---|---|---|---|---|---|---|
| 4. | 2021 [4] | Swin Transformer: Hierarchical Vision Transformer using Shifted Windows | Swin Transformer (Shifted Windows) | Accuracy, speed, scalability | Hierarchical design improves efficiency and accuracy in vision tasks | High computational demand for training |
| 5. | 2023 [5] | Facial Expression Recognition Algorithm Based on Swin Transformer | Swin Transformer for FER | Accuracy, F1 score | Swin-based FER improves feature extraction and recognition accuracy | Dataset limitations; high model complexity |
| 6. | 2024 [6] | Swin-FER: Swin Transformer for Facial Expression Recognition | Swin Transformer + optimized FER pipeline | Accuracy, robustness | Achieves high FER accuracy with improved attention modeling | Limited generalization across demographic variations |
| 7. | 2023 [7] | Facial Expression Recognition using Squeeze and Excitation-powered Swin Transformers | Swin Transformer + SE blocks | Accuracy, feature enhancement | SE blocks further enhance expressive feature extraction | Relatively high computational cost |

# LITERATURE SURVEY

| S.No | Year | Title | Method Used | Measures | Key Outcomes | Limitations |
|---|---|---|---|---|---|---|
| 8. | 2025 [8] | Mental health assessment model for college students based on facial expression recognition. | CNN + FERbased psychological assessment model | Accuracy, Evaluation metrics | FER-based mental health assessment improves early detection | FER-only; lacks physiological signals |
| 9. | 2024 [9] | Emotion Recognition of the Elderly Based on Transformer and Physiological Signals | Transformer + physiological features (HRV, ECG, GSR) | Accuracy, sensitivity | Multi-signal fusion improves emotion recognition for elderly users | Elderly-specific dataset; not generalizable |
| 10. | 2025 [10] | Emotional recognition and intervention of students in intelligent English teaching environment based on deep learning | Deep Learning (CNN, RNN, attention models) | Accuracy, intervention effectiveness | Deep learning improves emotion detection in smart English classrooms | Requires continuous monitoring; privacy concerns |

# LITERATURE SURVEY

| S.No | Year | Title | Method Used | Measures | Key Outcomes | Limitations |
|---|---|---|---|---|---|---|
| 11. | 2025 [11] | Emotion recognition for enhanced learning: using AI to detect students emotions and adjust teaching methods | AI + FER + adaptive teaching models | Accuracy, adaptability, usability | AI-based emotion detection supports personalized learning | Privacy, need for high quality video data |
| 12. | 2025 [12] | Deep learning based emotion recognition for analyzing students' psychological states during competitions | Deep Learning (CNN, Hybrid models) | Accuracy, robustness | Helps assess psychological states during high pressure events | Limited to competition settings |
| 13. | 2023 [13] | Real-Time Attention Monitoring System for Classroom: A Deep Learning Approach for Student's Behavior Recognition | Deep Learningbased behavior recognition | Accuracy, response time | Effective real-time monitoring of student attention | Hardware dependent; requires clear visuals |

# LITERATURE SURVEY

| S.No | Year | Title | Method Used | Measures | Key Outcomes | Limitations |
|---|---|---|---|---|---|---|
| 14. | 2022 [14] | Real-time mental stress detection using multimodality expressions with a deep learning framework | Multimodal DL (face + voice + physiological signals) | Accuracy, F1 score | High-accuracy stress detection using multimodal fusion | Requires synchronized multimodal inputs |

# MOTIVATION

- Students today face increasing academic pressure, competitive environments, social expectations, and uncertainty about their future. Continuous examinations, deadlines, online learning fatigue, and performance comparisons often affect students' emotional well-being.
- Many students experience stress or depressive tendencies but hesitate to seek help due to lack of awareness, fear of stigma, or difficulty in expressing their emotions openly. As a result, early warning signs of mental health issues often go unnoticed.
- Facial expressions naturally reflect a person's emotional state and are expressed unconsciously in daily activities. Emotions such as happiness, sadness, anger, fear, surprise, and neutrality provide valuable cues about an individual's mental condition. With advancements in deep learning and computer vision, these expressions can be analyzed accurately using camera-based systems.
- The motivation of this project is to develop a student-friendly mental health prediction system using Swin Transformer–based facial expression recognition by converting emotional patterns into a mental health score, the system enables self-assessment, early awareness, and timely guidance, helping students address mental health challenges before they become severe.

# STATISTICAL DATA

The proposed system uses facial expression datasets to train and evaluate the deep learning model for emotion recognition. The datasets provide labeled facial images that enable accurate learning of emotional patterns using transformer-based architectures.

**Datasets Used**:

- **FER2013 Dataset** contains approximately 35,887 grayscale facial images
- **Image size:** 48×48 pixels
- **Classes:** Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral.
- Divided into training, validation, and testing subsets.
- **CK+ (Cohn–Kanade Plus) Dataset** contains around 920 facial image sequences focuses on peak emotional expressions which are captured under controlled conditions to enhance recognition accuracy.

**Algorithms and Tools Used:**

- **Core Algorithm:** Swin Transformer for facial expression recognition supporting
- **Techniques:** Image preprocessing, normalization, and data augmentation
- **Framework:** PyTorch for model training and evaluation performance
- **Metrics:** Training loss and validation accuracy monitored through training logs

# PROBLEM STATEMENT

## Problem:

Rising stress levels are negatively affecting student's emotional well-being and academic performance, yet traditional assessment methods fail to detect early warning signs in real time, making early detection and timely support essential.

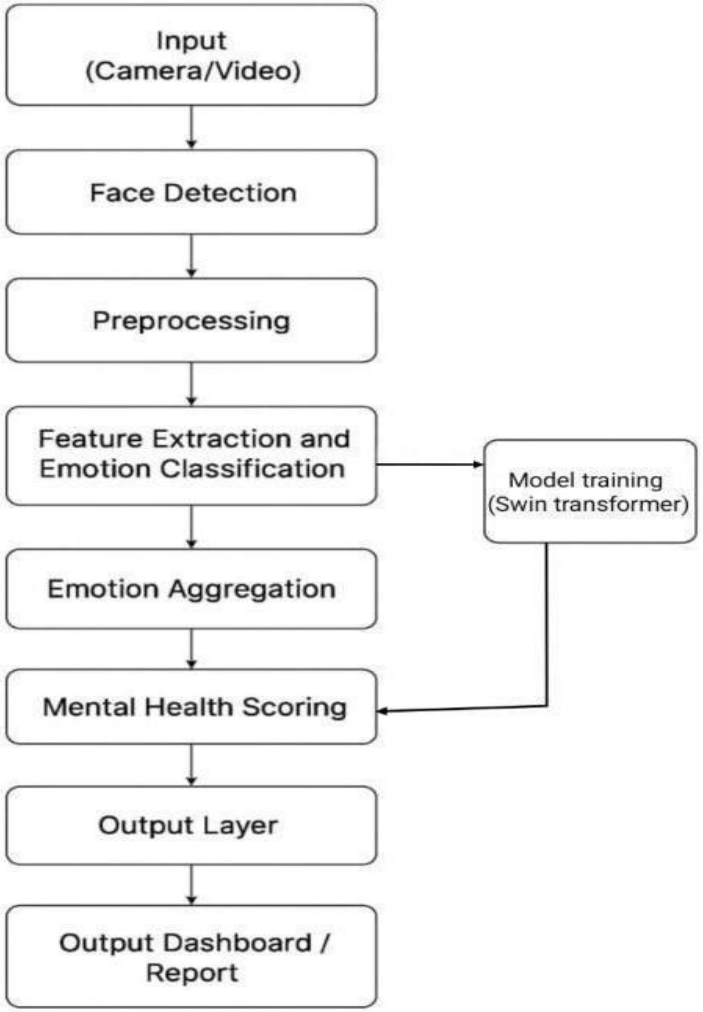
## Solution:

Stress and emotional imbalance among students have become a serious concern, affecting both academic performance and overall well-being. The proposed system detects stress through facial emotions and provides a mental health score in the output, helping users identify their condition early and encouraging timely self-assessment or seeking proper guidance.

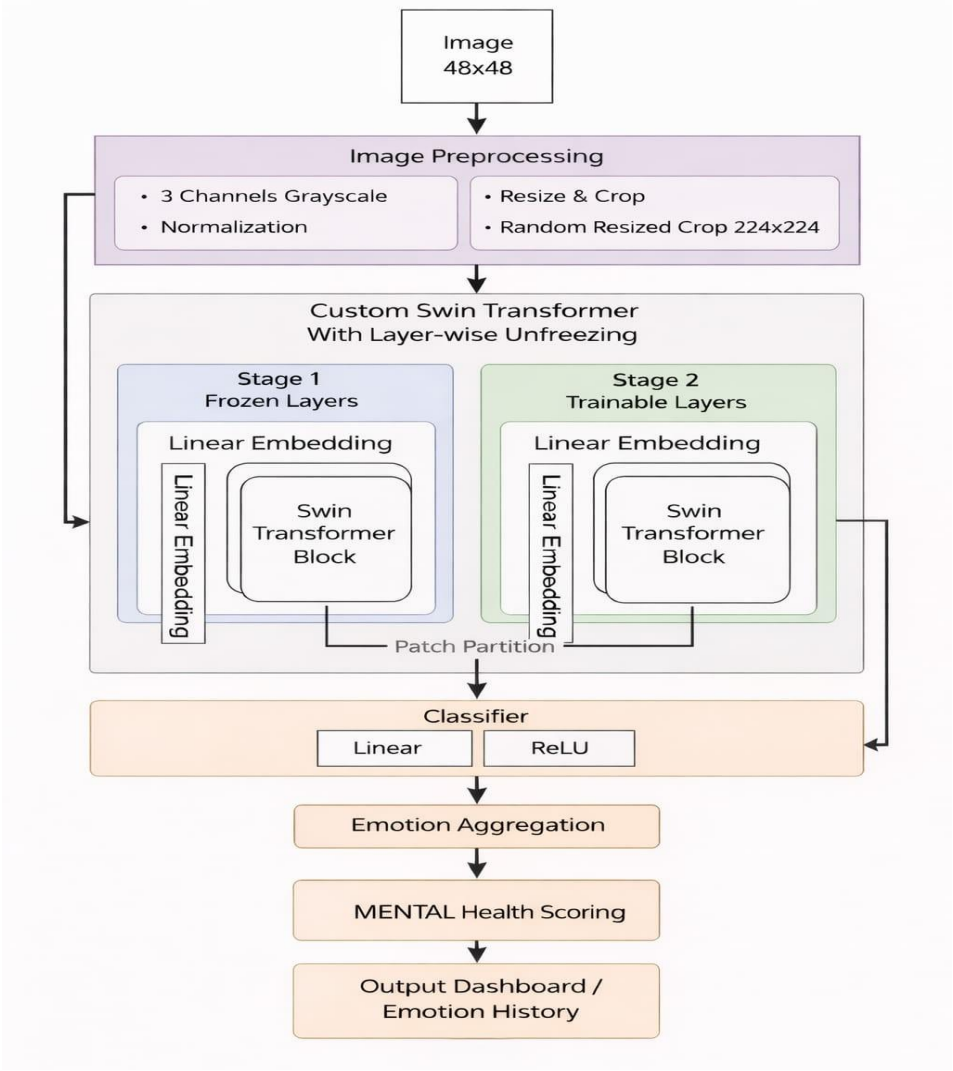
# OBJECTIVES

- To detect emotional states from facial expressions and use them as indicators to understand a person's mental well-being, with a focus on identifying early signs of stress among students.
- To analyze emotional patterns over time and convert them into a mental health score that helps users become aware of their emotional condition and decide whether to seek guidance or support.
- To build a reliable and student-friendly mental health monitoring approach that encourages timely self-assessment, reduces stigma, and motivates early intervention before problems worsen.

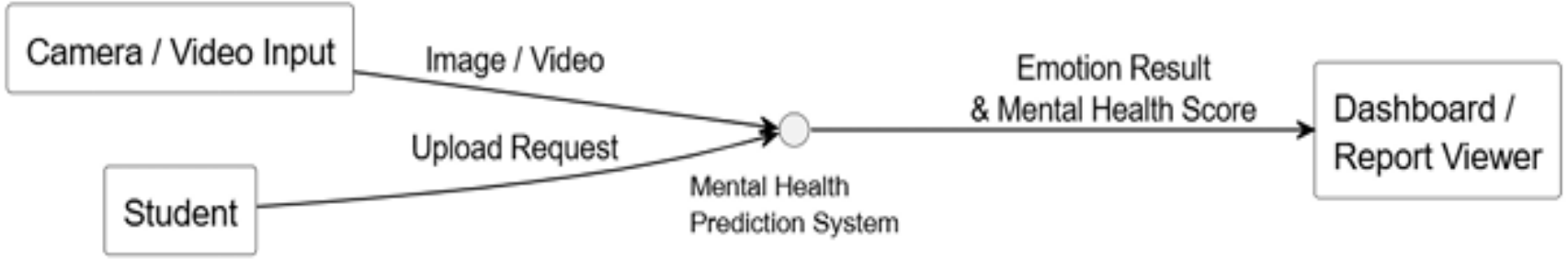
# ARCHITECTURE



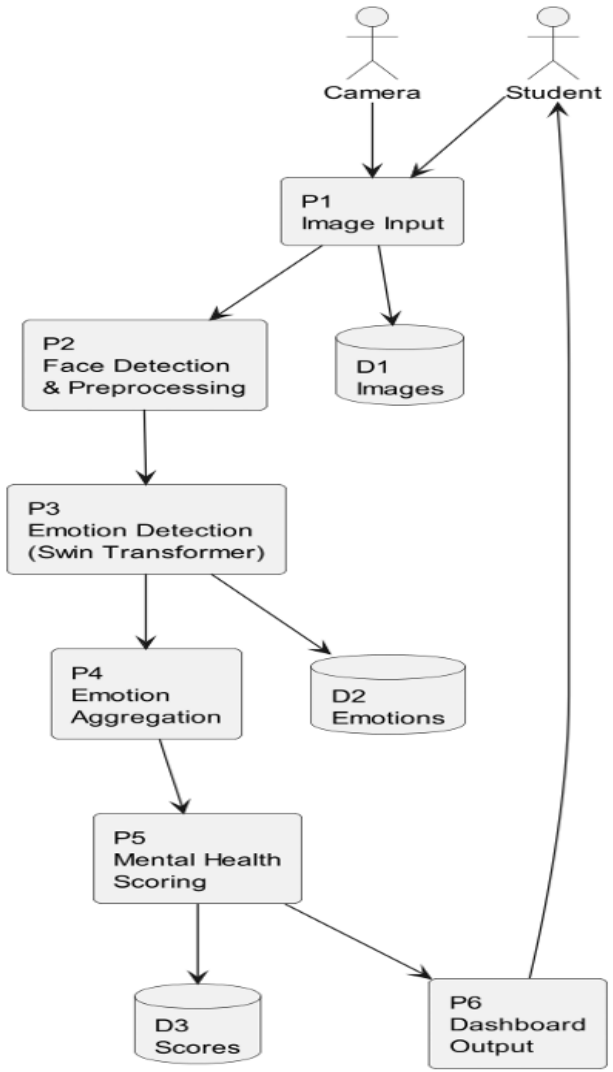
# FINALIZED ARCHITECTURE



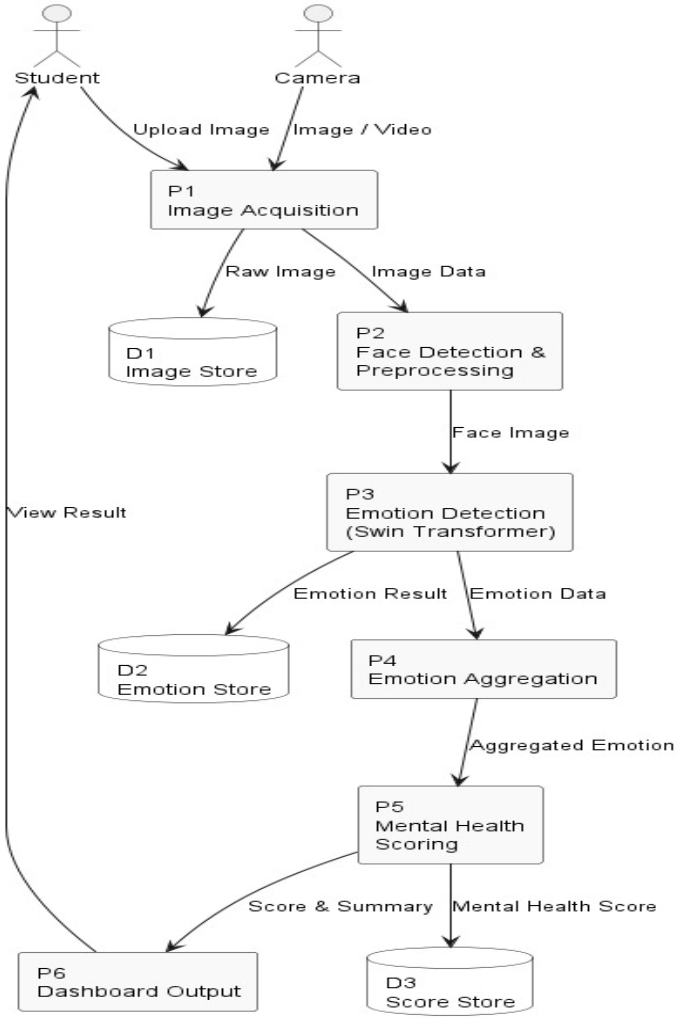
# Data Flow Diagram- level 0



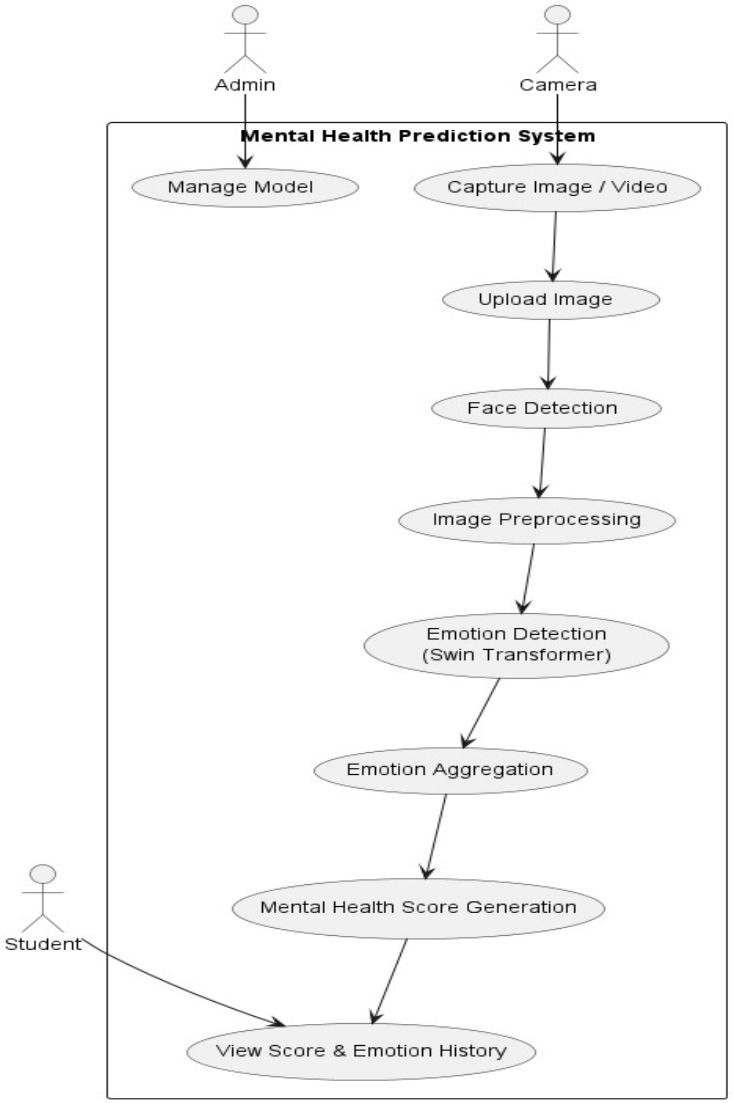
# Data Flow Diagram- level 1



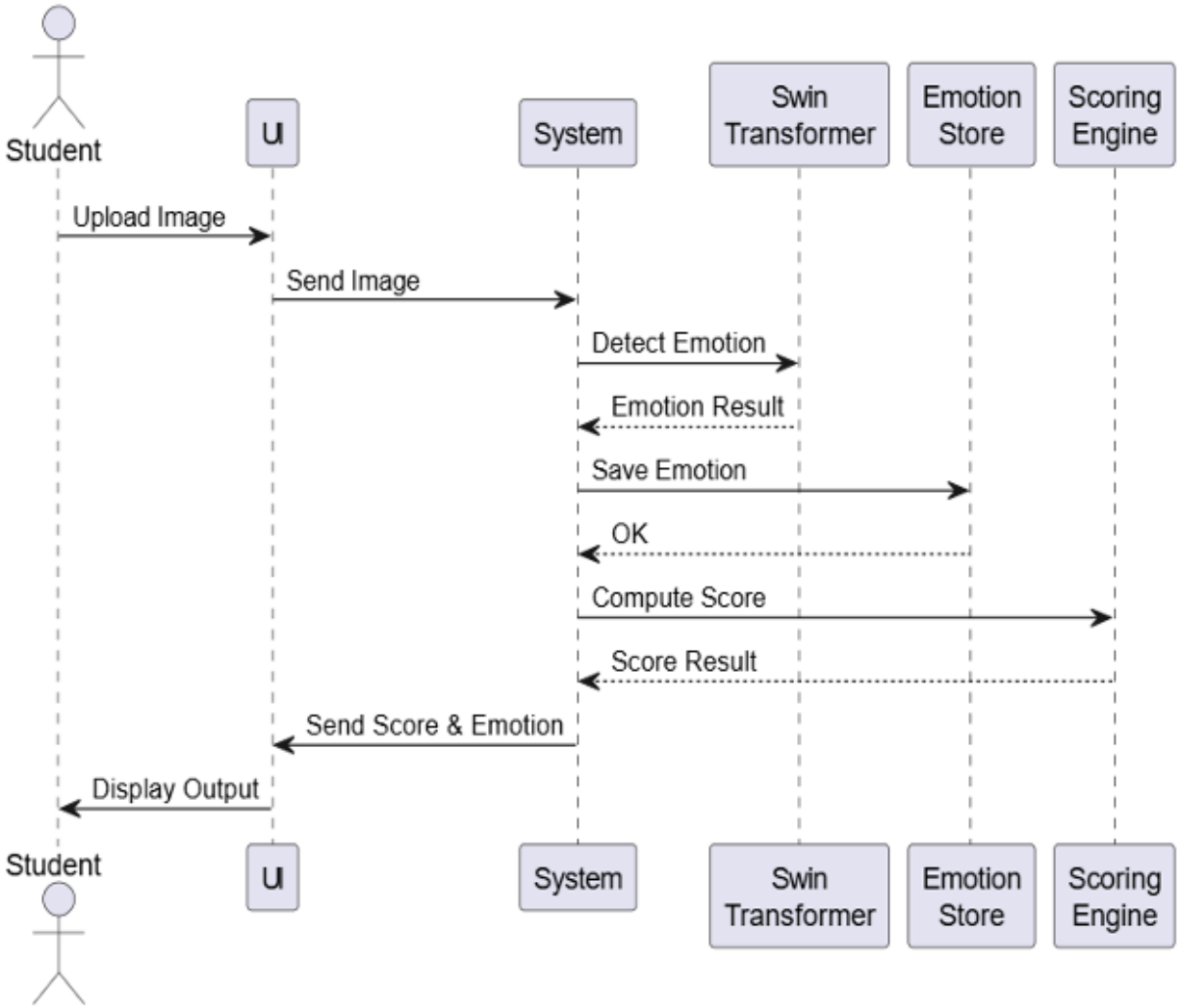
# Data Flow Diagram- level 2



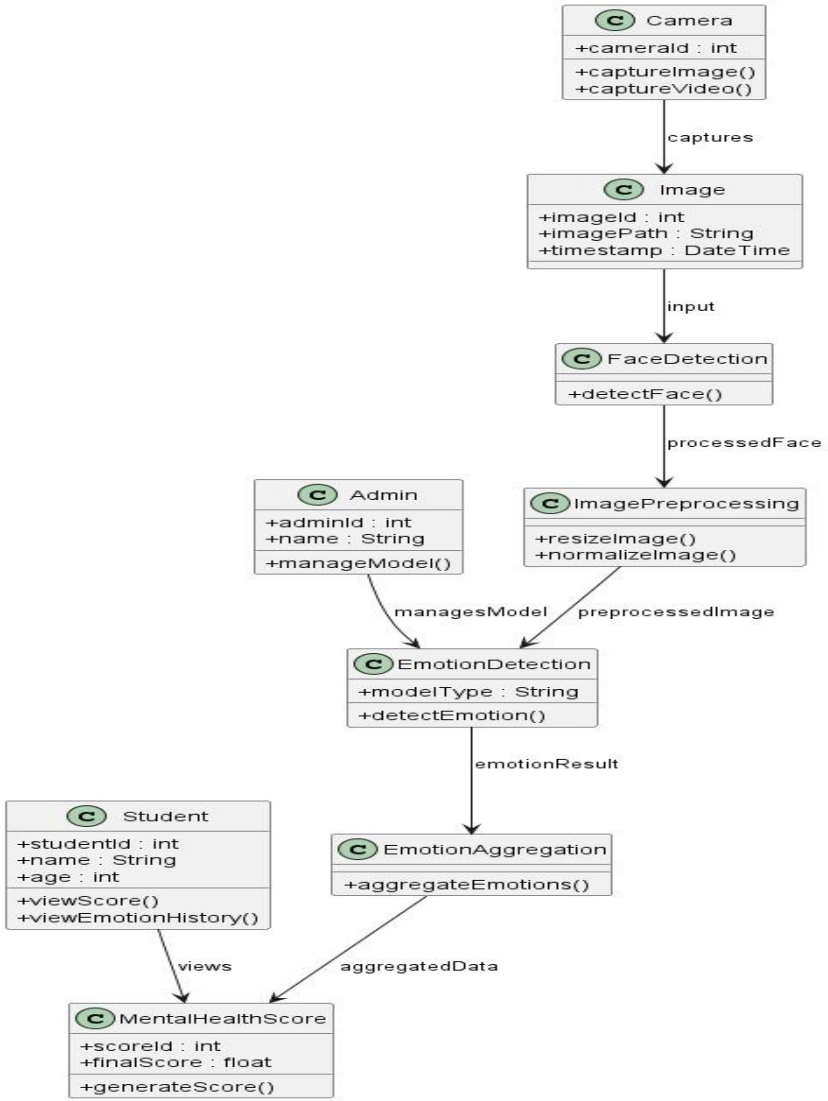
# Use case Diagram



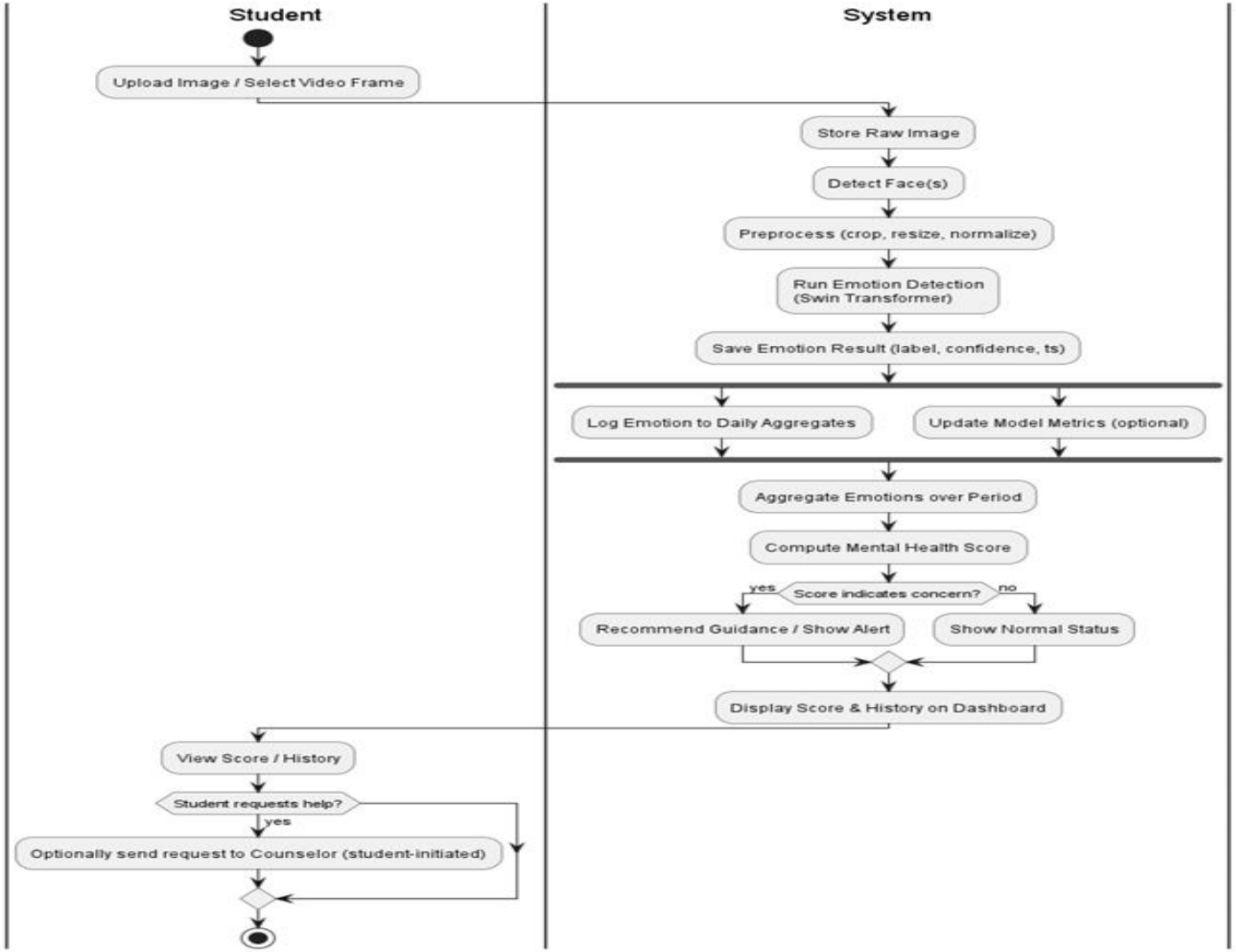
# Sequence Diagram



# Class diagram



# Activity Diagram



# PROCEDURE

- **Step 1:** Dataset Collection

Facial expression datasets such as FER2013 and CK+ are collected. These datasets contain labeled facial images representing emotions like Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

- **Step 2:** Data Preprocessing

All facial images are resized to the required input size (224×224) for the Swin Transformer. Pixel values are normalized, and data augmentation techniques such as flipping and rotation are applied to improve generalization.

- **Step 3:** Patch Partition and Embedding

Each input image is divided into fixed-size patches. These patches are flattened and converted into linear embeddings, which serve as the input tokens for the transformer model.

- **Step 4:** Swin Transformer-Based Feature Extraction

The Swin Transformer processes embedded patches using Window-based Multi-Head Self-Attention (W-MSA) Shifted Window Attention (SW-MSA) and Hierarchical feature representation. This enables efficient learning of both local and global facial features.

# PROCEDURE

- **Step 5:** Emotion Classification

The extracted features are passed through classification layers to predict one of the predefined emotion classes.

- **Step 6:** Emotion Aggregation

Predicted emotions are stored over time for each student. The frequency and dominance of emotions are analyzed to observe emotional trends.

- **Step 7:** Mental Health Score Generation

Based on emotional balance and recurring negative patterns, a Mental Health Score is calculated to support self-assessment and early awareness.

# IMPLEMENTATION

- The proposed system is implemented using Python as the primary programming language.
- PyTorch framework is used to develop and train the deep learning model.
- A Swin Transformer architecture is utilized for efficient emotion recognition.
- The system is trained on the FER2013 dataset, which contains labeled facial expressions.
- OpenCV is used for image acquisition, face detection, and preprocessing tasks.
- Input images or video frames are captured in real-time using a camera.
- Face detection is performed to identify the region of interest (ROI).
- Detected faces are resized to 224 × 224 pixels for model compatibility.
- Image normalization is applied to improve model performance and stability.
- The processed data is passed to the trained model for emotion analysis.

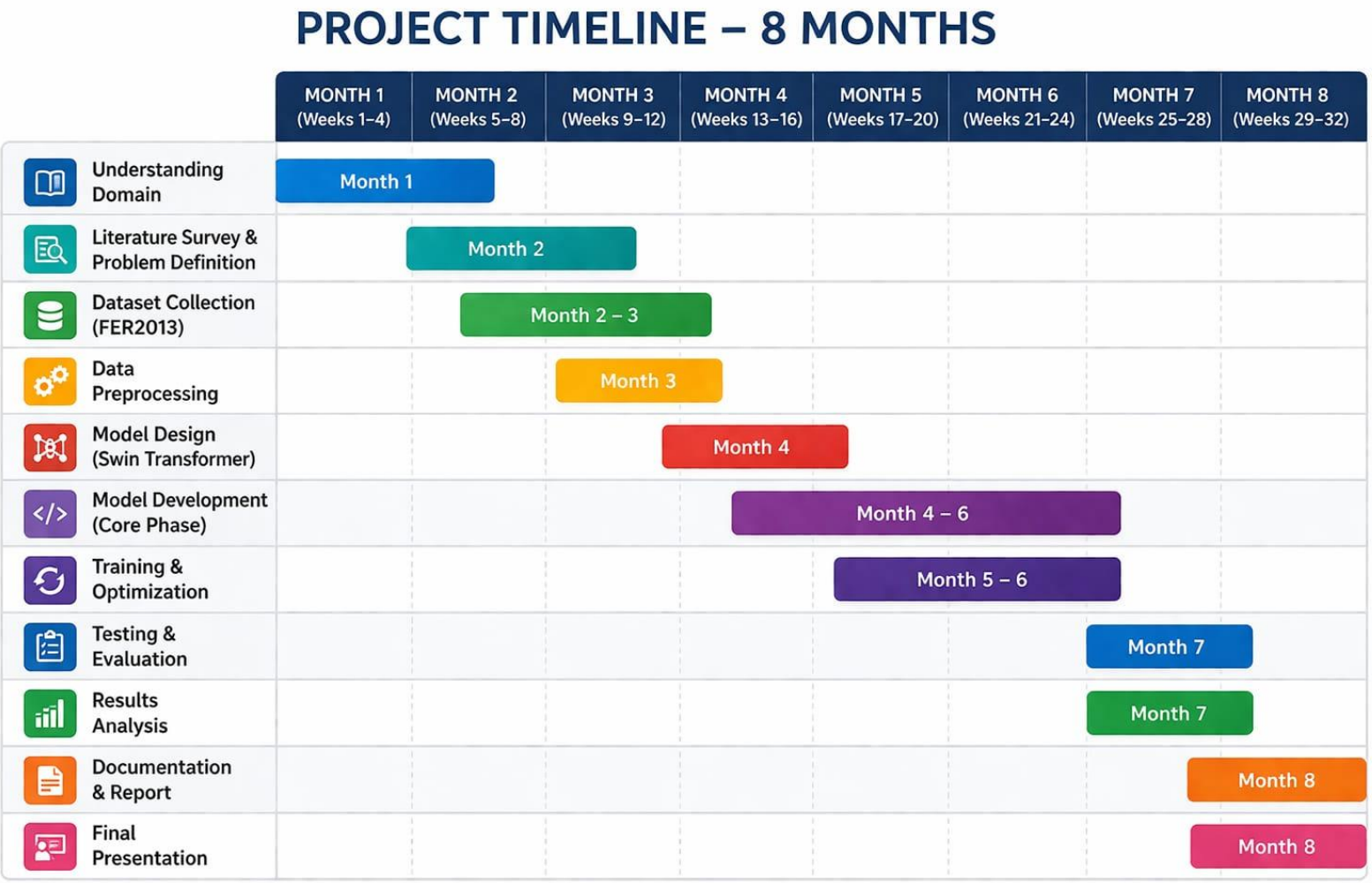
# IMPLEMENTATION

- The Swin Transformer extracts both local and global features from facial images.
- Attention mechanisms help the model focus on important facial regions.
- The extracted features are passed through fully connected layers.
- The model classifies input into seven emotions: happy, sad, angry, fear, surprise, disgust, and neutral.
- Pretrained weights are used to improve accuracy and reduce training time.
- The model is trained using the Cross Entropy Loss function for classification.
- AdamW optimizer is used for efficient weight updates and faster convergence.
- Training is performed in batches (batch size = 32) to handle large datasets.
- Multiple epochs are used to ensure better learning of patterns.
- The model performance improves gradually with reduced loss and increased accuracy.

# IMPLEMENTATION

- The system supports real-time emotion detection using webcam input.
- Each frame is processed and emotions are predicted continuously.
- Detected emotions are stored temporarily for further analysis.
- The system ensures smooth and fast processing for real-time performance.
- This enables continuous monitoring of user emotions.
- Emotion predictions are collected over a selected time interval (i.e, 20,30,45,60 seconds).
- Detected emotions are aggregated to analyze overall emotional behavior.
- A base mental health score (e.g., 50) is assigned initially.
- Positive emotions decrease the score, while negative emotions increase it.
- The final score represents the user's mental health condition and is displayed on the dashboard.

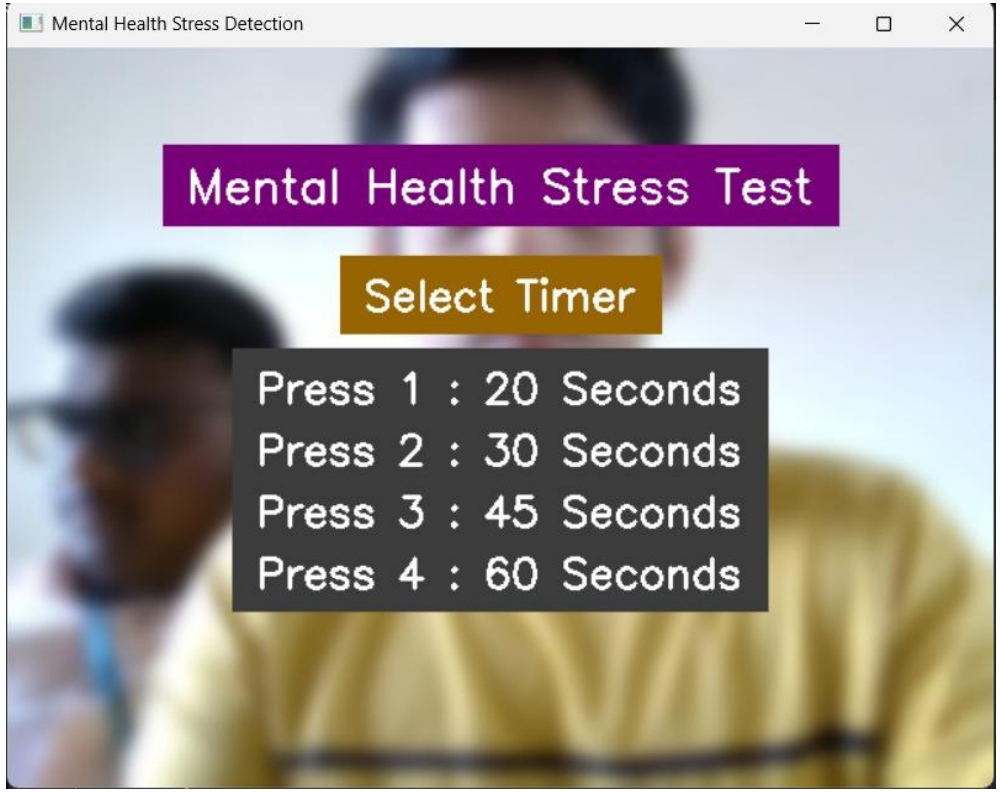
# TIMELINE

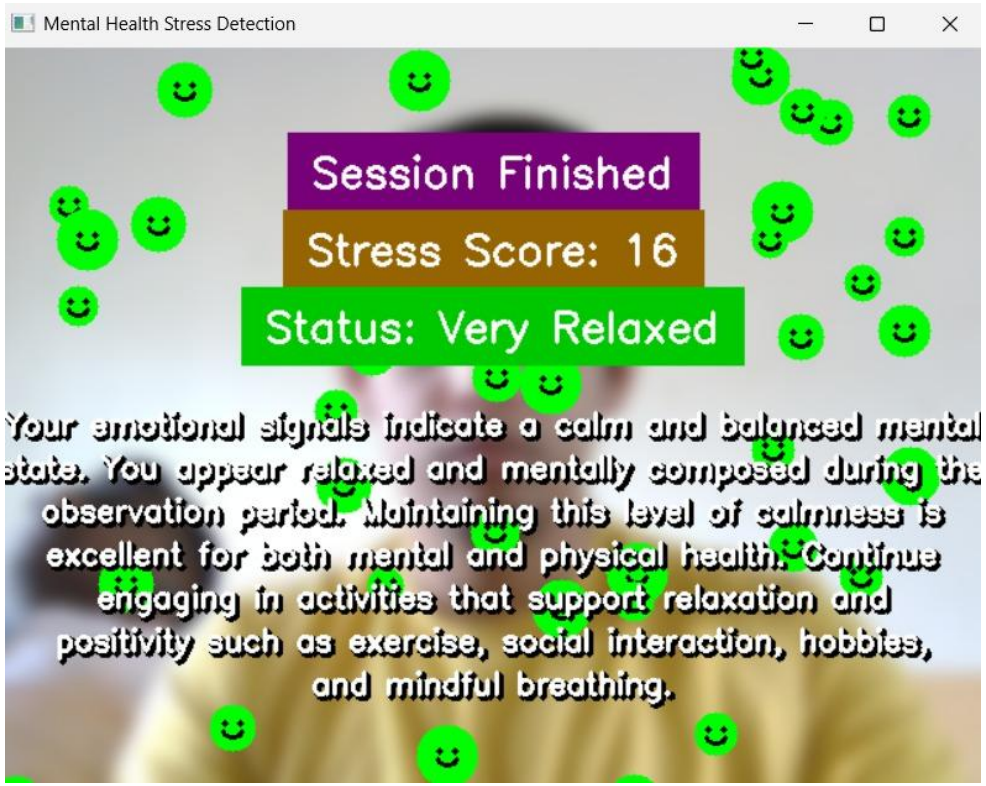

PROJECT TIMELINE – 8 MONTHS
MONTH 1 (Weeks 1–4)
MONTH 2 (Weeks 5–8)
MONTH 3 (Weeks 9–12)
MONTH 4 (Weeks 13–16)
MONTH 5 (Weeks 17–20)
MONTH 6 (Weeks 21–24)
MONTH 7 (Weeks 25–28)
MONTH 8 (Weeks 29–32)
Understanding Domain
Month 1
Literature Survey & Problem Definition
Month 2
Dataset Collection (FER2013)
Month 2 – 3
Data Preprocessing
Month 3
Model Design (Swin Transformer)
Month 4
Model Development (Core Phase)
Month 4 – 6
Training & Optimization
Month 5 – 6
Testing & Evaluation
Month 7
Results Analysis
Month 7
Documentation & Report
Month 8
Final Presentation
Month 8

# CONCLUSION

- The project promotes early mental health awareness by analyzing facial expressions to identify emotional patterns that may indicate stress or declining well-being
- It encourages students to reflect on their mental state using the generated mental health score and seek timely support from  a trusted adult, counselor, or professional if needed.
- This approach supports early intervention and can be strengthened in the future through multimodal inputs, expert collaboration, and strong ethical and privacy measures.

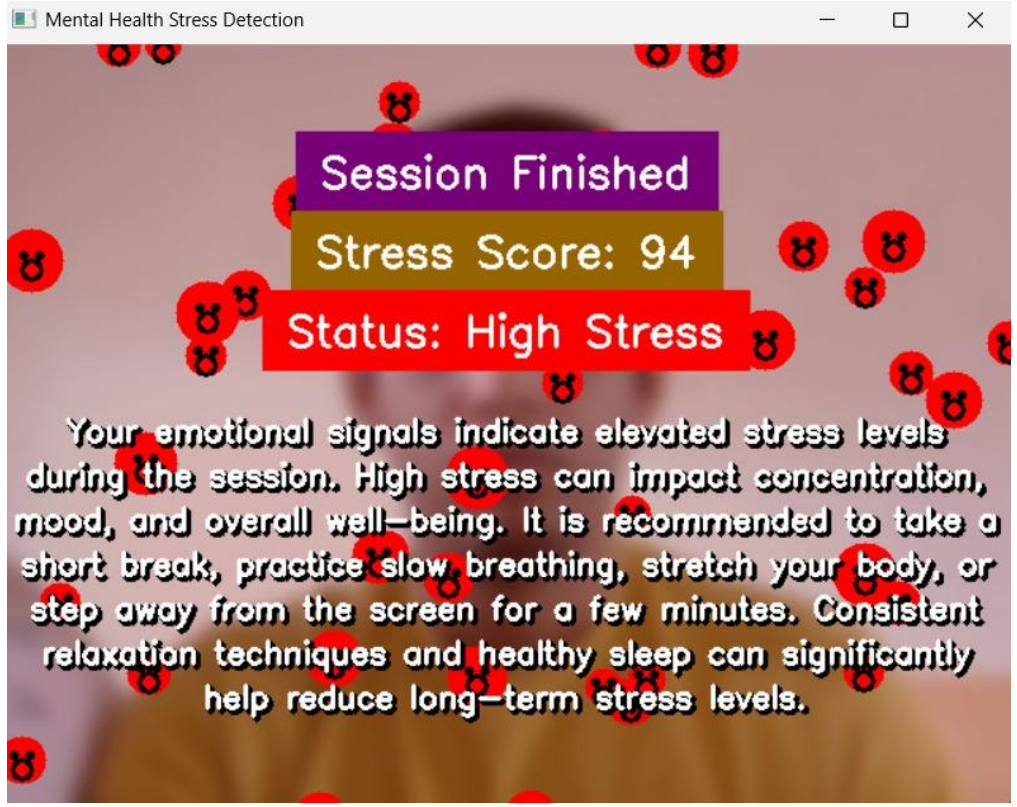
# RESULTS


Mental Health Stress Detection
Mental Health Stress Test
Select Timer
Press 1 : 20 Seconds
Press 2 : 30 Seconds
Press 3 : 45 Seconds
Press 4 : 60 Seconds


Mental Health Stress Detection
Session Finished
Stress Score: 16
Status: Very Relaxed
Your emotional signals indicate a calm and balanced mental state. You appear relaxed and mentally composed during the observation period. Maintaining this level of calmness is excellent for both mental and physical health. Continue engaging in activities that support relaxation and positivity such as exercise, social interaction, hobbies, and mindful breathing.

# RESULTS

Mental Health Stress Detection
Session Finished
Stress Score: 38
Status: Normal
Your emotional responses appear balanced with moderate fluctuations. This is considered a healthy and normal psychological state where the mind responds naturally to different stimuli. To maintain emotional balance, consider short breaks, deep breathing, staying hydrated, and maintaining positive social interactions throughout your day.


Mental Health Stress Detection
Session Finished
Stress Score: 94
Status: High Stress
Your emotional signals indicate elevated stress levels during the session. High stress can impact concentration, mood, and overall well-being. It is recommended to take a short break, practice slow breathing, stretch your body, or step away from the screen for a few minutes. Consistent relaxation techniques and healthy sleep can significantly help reduce long-term stress levels.

# RESULTS

```
100%|██████████| 113/113 [00:52<00:00,  2.14it/s]
Classification Report:

              precision    recall  f1-score   support

       Angry       0.64      0.58      0.61       491
     Disgust       0.67      0.76      0.71        55
        Fear       0.55      0.53      0.54       528
       Happy       0.92      0.86      0.89       879
     Neutral       0.63      0.73      0.68       626
         Sad       0.56      0.53      0.54       594
    Surprise       0.74      0.83      0.79       416

    accuracy                           0.69      3589
   macro avg       0.67      0.69      0.68      3589
weighted avg       0.69      0.69      0.69      3589
```

# SDG Mapping

| SDG | Mapped Indicator | SDG | Mapped Indicator | SDG | Mapped Indicator |
|---|---|---|---|---|---|
| 1 NO POVERTY | | 7 AFFORDABLE AND CLEAN ENERGY | | 13 CLIMATE ACTION | |
| 2 ZERO HUNGER | | 8 DECENT WORK AND ECONOMIC GROWTH | | 14 LIFE BELOW WATER | |
| 3 GOOD HEALTH AND WELL-BEING | 3.4.1, 3.4.2 | 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE | 9.5.1 | 15 LIFE ON LAND | |
| 4 QUALITY EDUCATION | 4.1.1, 4.a.1 | 10 REDUCED INEQUALITIES | 10.2.1 | 16 PEACE, JUSTICE AND STRONG INSTITUTIONS | |
| 5 GENDER EQUALITY | | 11 SUSTAINABLE CITIES AND COMMUNITIES | | 17 PARTNERSHIPS FOR THE GOALS | |
| 6 CLEAN WATER AND SANITATION | | 12 RESPONSIBLE CONSUMPTION AND PRODUCTION | | | |

# PUBLICATION

The International Journal of Versatile Research and Analysis (IJVRA) is a peer-reviewed, open-access multidisciplinary journal that publishes research in engineering, science, and technology. It provides global visibility through DOI indexing, follows a transparent review process, and enables researchers to share innovative ideas and findings with the academic community.

**DOI Link :** https://doi.org/10.56975/ijvra.v4i3.702333

# THANK YOU